

$$a) \int \frac{\sin^2 \sqrt{x}}{\sqrt{x}} dx = \left\{ \begin{array}{l} t = \sqrt{x} \\ dt = \frac{1}{2\sqrt{x}} dx \\ dx = 2\sqrt{x} dt \end{array} \right\} = \int \frac{\sin^2 t \cdot 2\sqrt{x}}{\sqrt{x}} dt =$$

$$1^o = 2 \int \sin^2 t dt = 2 \int \sin t \cdot \sin t dt = \left\{ \begin{array}{l} f = \sin t \quad f' = \cos t \\ g' = \sin t \quad g = -\cos t \end{array} \right\}$$

$$2^o = 2 \int \sin^2 t dt = 2 \int 1 \cdot \sin^2 t dt = \left\{ \begin{array}{l} f = \sin^2 t \quad f' = 2 \sin t \cos t = \sin 2t \\ g' = 1 \quad g = t \end{array} \right\}$$

$$= 2(t \sin^2 t - \int t \sin 2t dt) = \left\{ \begin{array}{l} f = t \quad f' = 1 \\ g' = \sin 2t \quad g = -\frac{1}{2} \cos 2t \end{array} \right\} =$$

$$= 2(t \sin^2 t + \frac{1}{2} t \cos 2t - \frac{1}{2} \int \cos 2t dt) =$$

$$= 2(t \sin^2 t + \frac{1}{2} t \cos 2t - \frac{1}{4} \sin 2t) + C =$$

$$= 2\sqrt{x} \sin^2 \sqrt{x} + \sqrt{x} \cos 2\sqrt{x} - \frac{1}{2} \sin 2\sqrt{x} + C$$

$$\begin{aligned} \cos 2x &= \cos^2 x - \sin^2 x \\ \cos 2x &= 1 - 2\sin^2 x \\ \sin^2 x &= \frac{1 - \cos 2x}{2} \end{aligned}$$

$$3^o \quad 2 \int \sin^2 t dt = 2 \int \frac{1 - \cos 2x}{2} dt = \int dt - \int \cos 2x dt =$$

$$= t - \frac{1}{2} \sin 2t + C$$

$$b) \int \frac{\operatorname{ctg} x}{\ln(\sin x)} dx = \int \frac{\cos x}{\ln(\sin x) \cdot \sin x} dx = \left\{ \begin{array}{l} t = \ln(\sin x) \\ dt = \frac{1}{\sin x} \cdot \cos x dx \\ dx = \frac{\sin x}{\cos x} \cdot dt \end{array} \right\} =$$

$$= \int \frac{1}{t} dt = \ln t + C = \ln(\ln(\sin x)) + C$$

$$c) \int \frac{x^5}{1+x^{12}} dx = \left\{ \begin{array}{l} t = x^6 \\ dt = 6x^5 dx \\ dx = \frac{1}{6x^5} dt \end{array} \right\} = \frac{1}{6} \int \frac{1}{1+t^2} dt = \frac{1}{6} \operatorname{arctg} t + C =$$

$$= \frac{1}{6} \operatorname{arctg} x^6 + C$$

$$d) \int \frac{\operatorname{tg} x}{1-\cos^3 x} dx = \int \frac{\sin x}{\cos x(1-\cos^3 x)} dx = \left\{ \begin{array}{l} t = \cos x \\ dt = -\sin x dx \\ dx = \frac{-dt}{\sin x} \\ \int \frac{\sin x}{t(1-t^3)} \cdot \frac{(-dt)}{\sin x} \end{array} \right\} =$$

$$= - \int \frac{1}{t(1-t^3)} dt = - \int \frac{1}{t(1-t)(1+t+t^2)} dt =$$

$$= - \int \frac{a}{t} + \frac{b}{1-t} + \frac{ct+d}{1+t+t^2} dt$$

$$\frac{1}{t(1-t^3)} = \frac{a}{t} + \frac{b}{1-t} + \frac{ct+d}{1+t+t^2}$$

$$\frac{1}{t(1-t^3)} = \frac{a(1-t^3) + b t(1+t+t^2) + (t-t^2)(ct+d)}{t(1-t)(1+t+t^2)}$$

$$\frac{1}{t(1-t^3)} = \frac{a - at^3 + bt + bt^2 + bt^3 + ct^2 + dt - ct^3 - dt^2}{t(1-t^3)} = \frac{(b-a-c)t^3 + (b+c-d)t^2 + (b+d)ta}{t(1-t^3)}$$

$$b-a-c=0$$

$$b+c-d=0$$

$$b+d=0$$

$$a=1$$

$$a=1$$

$$a=1$$

$$b=1$$

$$c=0$$

$$b-c=1$$

$$2c-d=1$$

$$b+d=0$$

$$-d-c=1$$

$$c=-d-1$$

$$-2d-2-d=1$$

$$-3d=3$$

$$d=-1$$

$$= - \int \frac{1}{t} + \frac{1}{1-t} + \frac{-1}{1+t+t^2} dt = - \int \frac{1}{t} dt - \int \frac{1}{1-t} dt + \int \frac{1}{1+t+t^2} dt =$$

$$= -\ln t + \ln(1-t) - \ln(1+t+t^2) \cdot \frac{1}{2t+1} + C$$

$$e) \int \frac{\ln(2x+1)}{x^3} dx = \left\{ \begin{array}{l} f = \ln(2x+1) \quad f' = \frac{2}{2x+1} \\ g' = x^{-3} \quad g = -\frac{1}{2} x^{-2} \end{array} \right\} =$$

$$= -\frac{\ln(2x+1)}{2x^2} + \int \frac{1}{x^2(2x+1)} dx$$

$$\int \frac{1}{x^2(2x+1)} dx = \int \frac{a}{x} + \frac{b}{x^2} + \frac{c}{2x+1} dx = \int \frac{-2}{x} + \frac{1}{x^2} + \frac{4}{2x+1} dx$$

$$\frac{1}{x^2(2x+1)} = \frac{(ax+b)(2x+1) + cx^2}{x^2(2x+1)} = -2 \int \frac{1}{x} dx + \int \frac{1}{x^2} dx + 4 \int \frac{1}{2x+1} dx$$

$$1 = 2ax^2 + ax + 2bx + b + cx^2 = -2\ln x + \frac{1}{2x} \ln x^2 + 2\ln(2x+1)$$

$$1 = (2a+c)x^2 + (2b+a)x + b$$

$$2a+c=0 \rightarrow -4+c=0 \rightarrow c=4$$

$$2b+a=0 \rightarrow 2+a=0 \rightarrow a=-2$$

$$b=1$$

$$f) \int x^2 \ln \frac{1+x}{1-x} dx = \left\{ \begin{array}{l} f = \ln \frac{1+x}{1-x} \quad f' = \frac{1-x}{1+x} \cdot \frac{1-x+(1+x)}{(1-x)^2} = \frac{2}{(1+x)(1-x)} \\ g' = x^2 \quad g = \frac{1}{3}x^3 \quad f' = \frac{2}{1-x^2} \end{array} \right\} =$$

$$= \frac{1}{3}x^3 \ln \frac{1+x}{1-x} - \frac{2}{3} \int \frac{x^3}{1-x^2} dx =$$

$$= A - \frac{2}{3} \int -x + \frac{x}{1-x^2} dx =$$

$$= A + \frac{2}{3} \int x dx - \frac{2}{3} \int \frac{x}{(1-x)(1+x)} dx =$$

$$= A + \frac{1}{3}x^2 - \frac{2}{3} \int \frac{1}{2} \cdot \frac{1}{1-x} - \frac{1}{2} \cdot \frac{1}{1+x} dx =$$

$$= A + \frac{1}{3}x^2 - \frac{1}{3} \int \frac{1}{1-x} dx + \frac{1}{3} \int \frac{1}{1+x} dx =$$

$$= A + \frac{1}{3}x^2 + \frac{1}{3} \ln(1-x) + \frac{1}{3} \ln(1+x) + C$$

$$\frac{-x}{x^3 : (-x^2 + 1)} = \frac{-x^3 + x}{x}$$

$$\frac{x}{(1-x)(1+x)} = \frac{a}{1-x} + \frac{b}{1+x}$$

$$x = a + ax + b - bx$$

$$x = (a-b)x + a+b$$

$$a-b=1 \quad a=\frac{1}{2}$$

$$a+b=0$$

$$1+2b=0$$

$$b=-\frac{1}{2}$$

$$\begin{aligned}
 \text{g) } \int \sin 2x \cdot e^{-\cos x} dx &= 2 \int \sin x \cdot \cos x \cdot e^{-\cos x} dx = \left\{ \begin{array}{l} t = -\cos x \\ dt = \sin x dx \\ dx = \frac{dt}{\sin x} \end{array} \right\} = \\
 &= -2 \int t \cdot e^t dt = \left\{ \begin{array}{l} f = t \quad f' = 1 \\ g' = e^t \quad g = e^t \end{array} \right\} =
 \end{aligned}$$

$$= -2te^t - 2 \int e^t dt = -2te^t - 2e^t = 2\cos x e^{-\cos x} - 2e^{-\cos x} + C$$

$$\begin{aligned}
 \int \frac{1+e^x}{e^x - e^{2x}} dx &= \left\{ \begin{array}{l} e^x = t \\ e^x dx = dt \\ \frac{1+t}{t-t^2} \cdot \frac{dt}{t} \end{array} \right\} = \int \frac{1+t}{t^2-t^3} dt = \int \frac{1+t}{t^2(1-t)} dt = \\
 &= \int \frac{a}{t} + \frac{b}{t^2} + \frac{c}{1-t}
 \end{aligned}$$